L1 Trigger matched cross-section results

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Outline

1. Events from Runs 2 and 3 (Input Files Used)
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3. LH2 Cross-Section Comparison (With/Without L1 Trigger Check)
4. Cross-Section comparison to RS67
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7. Cross-Section Ratios to RS67 (xF bins)
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Events from runs 2 and 3 (Input Files Used)

Note: Currently using all the runs saved in runs 2 and 3.

Harsha's ROOT files:
/seaquest/users/harshaka/e906_project/e906-root-ana/work_gpvm/scripts/results_runFinalTree/merged_results_e906_mixing/:

- merged_RS57_LH2_1_1138.root
- merged_RS57_LD2_3_1138.root
- merged_RS57_Empty_2_1138.root
- merged_RS59_LH2_1_465.root
- merged_RS59_LD2_3_466.root
- merged_RS59_Empty_2_466.root
- merged_RS62_LH2_1_1234.root
- merged_RS62_LD2_3_1237.root
- merged_RS62_Empty_2_1234.root
- merged_RS70_LH2_1_264.root
- merged_RS70_LD2_3_266.root
- merged_RS70_Empty_2_267.root

Abi's ROOT files:
/seaquest/users/apun/e906_projects/rs67_merged_files/:

- merged_RS67_3089LH2.root
- merged_RS67_3089LD2.root
- merged_RS67_3089flask.root

Event Selection Applied

Chuck Cuts (DocDB2111-V42**)** with beam_offset $= 1.6$ cm and mass cut $= 4.2$ GeV/c^2:

Dimuon Cuts (dimuon_cut_2111_v42)

- Vertex: $|dx| < 0.25$, $|dy - 1.6| < 0.22$, $-280 < dz < -5$ cm
- Transverse Vertex: $dx^2 + (dy - 1.6)^2 < 0.06$ cm^2
- Momentum: $|dp_x| < 1.8$, $|dp_y| < 2.0$, $dp_x^2 + dp_y^2 < 5.0$ (GeV/c)^2, $38 < dp_z < 116$ GeV/c
- Kinematics: $4.2 <$ mass < 8.8 GeV/c^2, $-0.1 < x_F < 0.95$, $0.05 < x_T \leq 0.58$, $|\cos\theta| < 0.5$
- Fit & Separation: $|$trackSeparation$| < 270$ cm, $\chi^2_{\text{dimuon}} < 18$

Single Track Cuts (track1_cut & track2_cut, $i = 1, 2$)

- Target/Dump χ^2: $\chi^2_{i,\text{target}} < 15$, $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{upstream}}$, $\chi^2_{i,\text{target}} < 1.5\chi^2_{i,\text{dump}}$
- Track Quality: nHits$_i > 13$, $\chi^2_i/(\text{nHits}_i - 5) < 12$, $-320 < z_{i,v} < -5$ cm
- Apertures: $x_{i,t}^2 + (y_{i,t} - 1.6)^2 < 320$, $16 < x_{i,d}^2 + (y_{i,d} - 1.6)^2 < 1100$ cm^2
- Station 1 & 3: $9 < p^{\text{st1}}_{z,i} < 75$, $|p^{\text{st1}}_{y,i}| > 0.02$ GeV/c, $y_{i,\text{st1}}/y_{i,\text{st3}} < 1$, $y_{i,\text{st1}} \cdot y_{i,\text{st3}} > 0$
- Δp Matching: $||p^{\text{st1}}_{x,i} - p^{\text{st3}}_{x,i}|0.416| < 0.008$, $|p^{\text{st1}}_{y,i} - p^{\text{st3}}_{y,i}| < 0.008$,

Track Pair Cuts (tracks_cut_2111_v42)

- $|\chi^2_{1,\text{target}} + \chi^2_{2,\text{target}} - \chi^2_{\text{dimuon}}| < 2.0$
- $y_{1,\text{st3}} \cdot y_{2,\text{st3}} < 0$ (opposite top/bottom halves, no beam offset)
- nHits$_1$ + nHits$_2 >$ 29 and nHits1St1 + nHits2St1 > 8
- $|x_{1,\text{st1}} + x_{2,\text{st1}}| < 42$ cm

Occupancy Cuts (occ_cut_2111_v42)

- $D_1 < 400$, $D_2 < 400$, $D_3 < 400$, and $D_1 + D_2 + D_3 < 1000$
- ***Note*: Intensity cuts (**intensity_cut_2111_v42**) are not applied.**

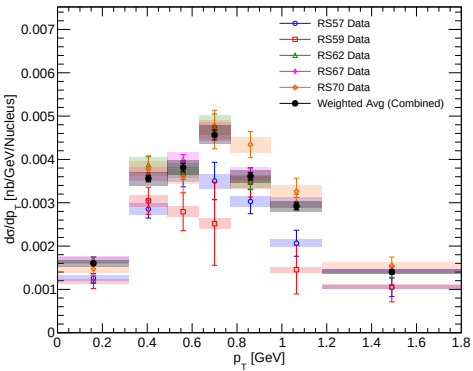
Additional Cuts Applied

- Kinematic Fiducial Cut: $(4.2 <$ mass < 8.8 GeV/c^2) & ($0.0 < x_F < 0.8$) & ($0 < p_T < 1.8$ GeV/c)
- L1 Trigger Match: **Track-level L1 trigger check for** roadID

LH2 Cross-Section Comparison All Roadsets (with/without L1 trigger Check)

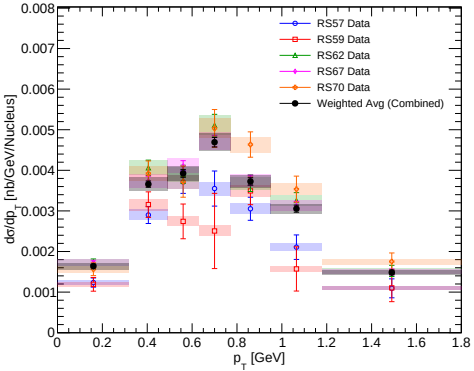
With L1 Trigger Check

Drell-Yan Absolute Cross-Section (LH2)



Without L1 Trigger

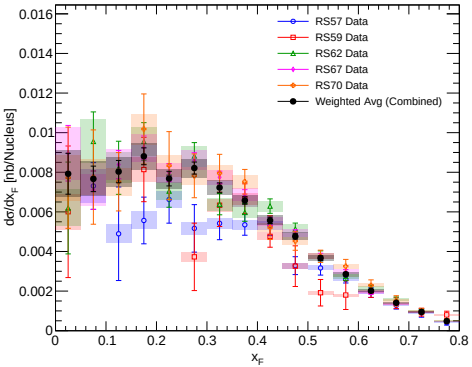
Drell-Yan Absolute Cross-Section (LH2)



LH2 Cross-Section Comparison All Roadsets - xF bins (with/without L1 trigger Check)

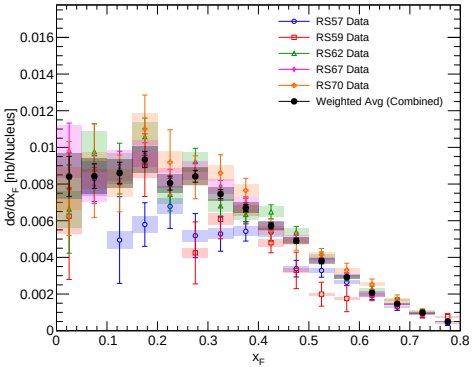
With L1 Trigger Check

Drell-Yan Absolute Cross-Section (LH2)



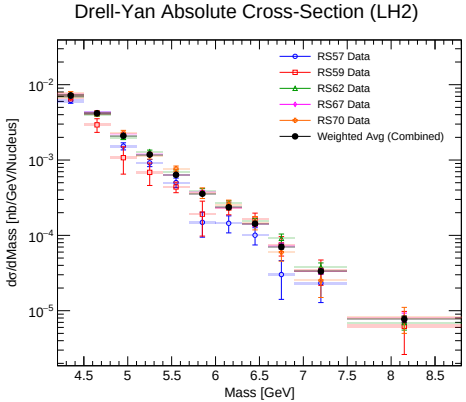
Without L1 Trigger

Drell-Yan Absolute Cross-Section (LH2)

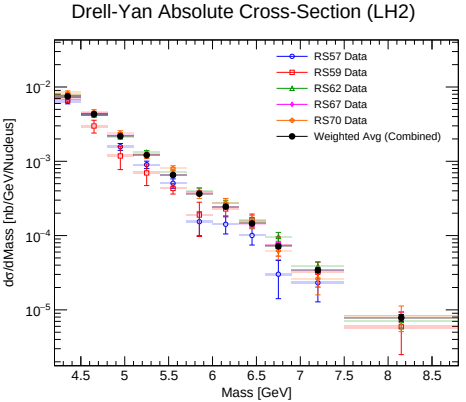


LH2 Cross-Section Comparison All Roadsets - mass bins (with/without L1 trigger Check)

With L1 Trigger Check



Without L1 Trigger



Cross-Section comparison to RS67

With L1 trigger check

Table 1: Best-fit single differential cross-section ratios vs. pT (RS/RS67) for LH_2 and LD_2 targets.

Roadset Ratio	LH_2 Best Fit ($\pm$ Error)	LD_2 Best Fit ($\pm$ Error)
RS57 / RS67	0.785 ± 0.037	0.739 ± 0.022
RS59 / RS67	0.761 ± 0.051	0.730 ± 0.036
RS62 / RS67	0.993 ± 0.034	0.952 ± 0.023
RS70 / RS67	1.016 ± 0.044	0.975 ± 0.031
RS57-70 / RS67	0.971 ± 0.027	0.942 ± 0.018

Without L1 trigger check

Table 2: Best-fit single differential cross-section ratios vs. pT (RS/RS67) for LH_2 and LD_2 targets.

Roadset Ratio	LH_2 Best Fit ($\pm$ Error)	LD_2 Best Fit ($\pm$ Error)
RS57 / RS67	0.773 ± 0.036	0.736 ± 0.022
RS59 / RS67	0.748 ± 0.049	0.683 ± 0.034
RS62 / RS67	1.020 ± 0.035	0.981 ± 0.023
RS70 / RS67	1.053 ± 0.046	0.991 ± 0.031
RS57-70 / RS67	0.976 ± 0.027	0.945 ± 0.018

Cross-Section comparison to RS67 (xF bins)

With L1 trigger check

Table 3: Best-fit single differential cross-section ratios vs. xF (RS/RS67) for LH_2 and LD_2 targets.

Roadset Ratio	LH_2 Best Fit ($\pm$ Error)	LD_2 Best Fit ($\pm$ Error)
RS57 / RS67	0.808 ± 0.034	0.746 ± 0.021
RS59 / RS67	0.834 ± 0.051	0.734 ± 0.036
RS62 / RS67	0.996 ± 0.030	0.955 ± 0.021
RS70 / RS67	1.016 ± 0.040	0.958 ± 0.029
RS57-70 / RS67	0.980 ± 0.023	0.946 ± 0.016

Without L1 trigger check

Table 4: Best-fit single differential cross-section ratios vs. xF (RS/RS67) for LH_2 and LD_2 targets.

Roadset Ratio	LH_2 Best Fit ($\pm$ Error)	LD_2 Best Fit ($\pm$ Error)
RS57 / RS67	0.804 ± 0.034	0.745 ± 0.021
RS59 / RS67	0.820 ± 0.050	0.691 ± 0.034
RS62 / RS67	1.028 ± 0.031	0.986 ± 0.022
RS70 / RS67	1.056 ± 0.042	0.976 ± 0.030
RS57-70 / RS67	0.987 ± 0.023	0.950 ± 0.016

Cross-Section comparison to RS67 (mass bins)

With L1 trigger check

Table 5: Best-fit single differential cross-section ratios vs. Mass (RS/RS67) for LH_2 and LD_2 targets.

Roadset Ratio	LH_2 Best Fit ($\pm$ Error)	LD_2 Best Fit ($\pm$ Error)
RS57 / RS67	0.761 ± 0.037	0.722 ± 0.021
RS59 / RS67	0.771 ± 0.055	0.704 ± 0.037
RS62 / RS67	1.010 ± 0.030	0.948 ± 0.020
RS70 / RS67	1.018 ± 0.041	0.972 ± 0.029
RS57-70 / RS67	0.976 ± 0.021	0.937 ± 0.015

Without L1 trigger check

Table 6: Best-fit single differential cross-section ratios vs. Mass (RS/RS67) for LH_2 and LD_2 targets.

Roadset Ratio	LH_2 Best Fit ($\pm$ Error)	LD_2 Best Fit ($\pm$ Error)
RS57 / RS67	0.743 ± 0.037	0.713 ± 0.021
RS59 / RS67	0.750 ± 0.053	0.661 ± 0.034
RS62 / RS67	1.033 ± 0.031	0.976 ± 0.021
RS70 / RS67	1.052 ± 0.042	0.984 ± 0.029
RS57-70 / RS67	0.978 ± 0.021	0.938 ± 0.015

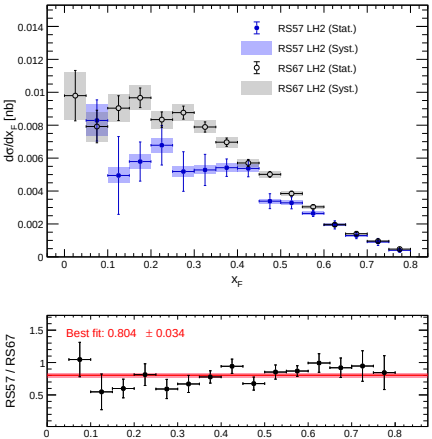
Next Steps

- There is no significant difference between the cross-section results with and without L1 trigger check. But we need to integrate the check to ensure the accurate cross-section calculations.

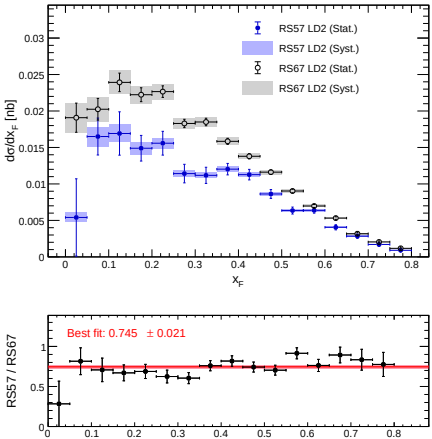
Backup Slides

Cross-Section Ratio RS57 to RS67 (xF bins)

LH2 (RS57 vs RS67)

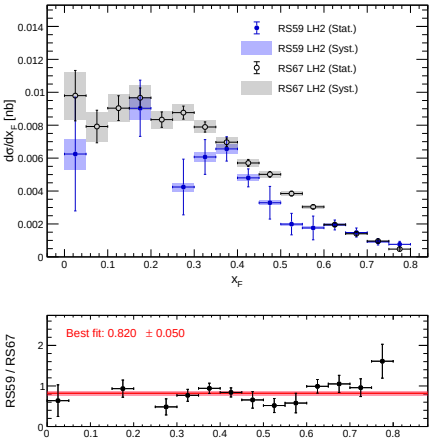


LD2 (RS57 vs RS67)

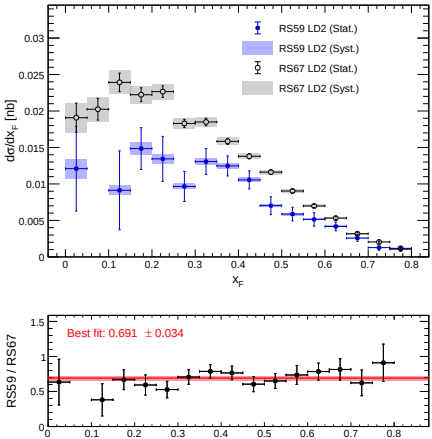


Cross-Section Ratio RS59 to RS67 (xF bins)

LH2 (RS59 vs RS67)

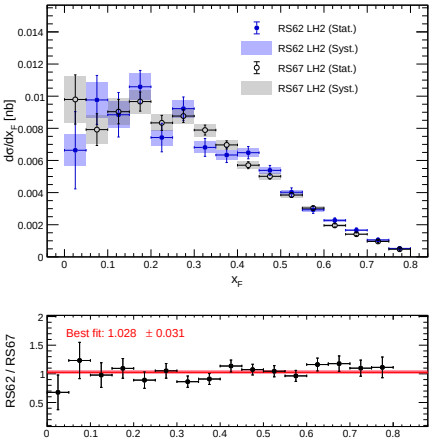


LD2 (RS59 vs RS67)

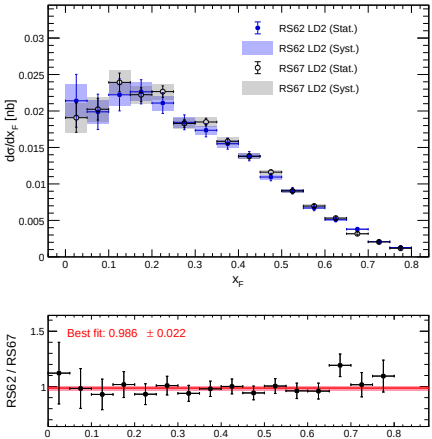


Cross-Section Ratio RS62 to RS67 (xF bins)

LH2 (RS62 vs RS67)

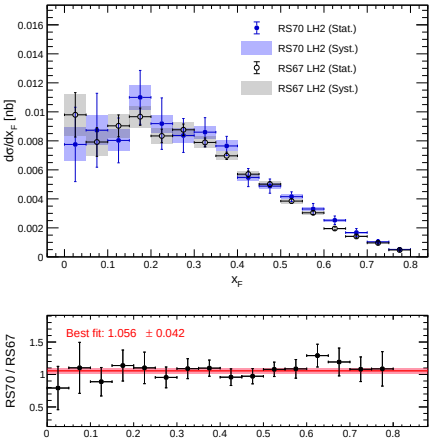


LD2 (RS62 vs RS67)

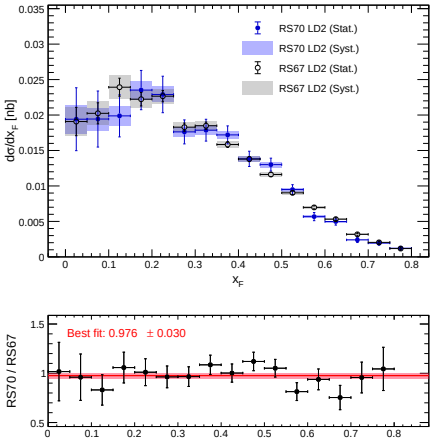


Cross-Section Ratio RS70 to RS67 (xF bins)

LH2 (RS70 vs RS67)

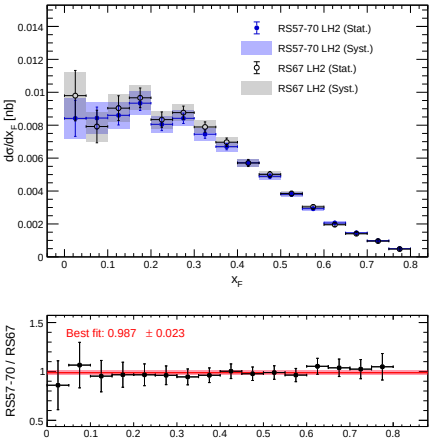


LD2 (RS70 vs RS67)

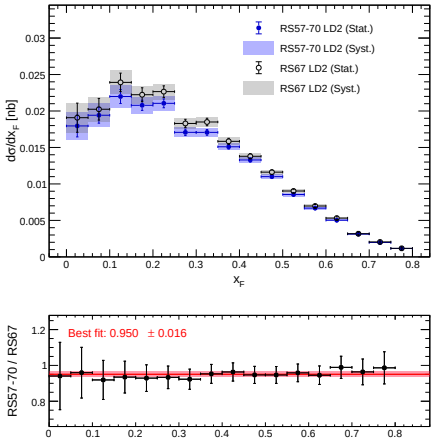


Cross-Section Ratio RS57-70 to RS67 (xF bins)

LH2 (RS57-70 vs RS67)

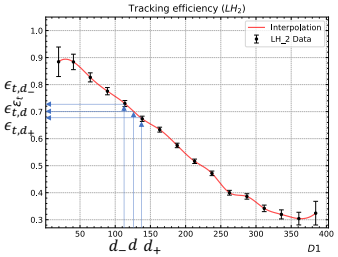


LD2 (RS57-70 vs RS67)

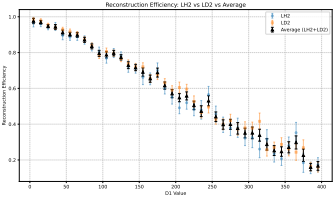


New Reconstruction Efficiency Correction (D1-10)

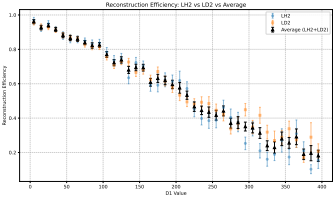
Previous Global Curve (All RS)



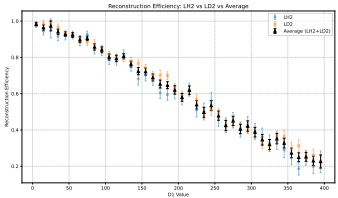
RS57 Efficiency Curve



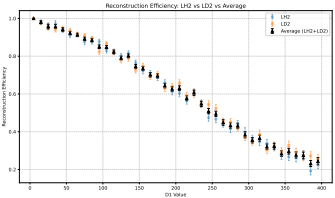
RS59 Efficiency Curve



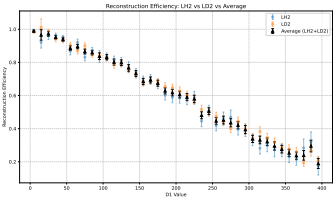
RS62 Efficiency Curve



RS67 Efficiency Curve

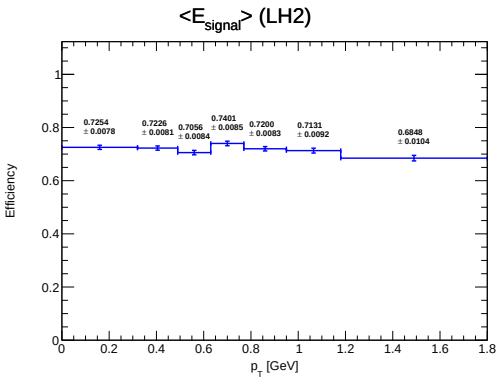


RS70 Efficiency Curve

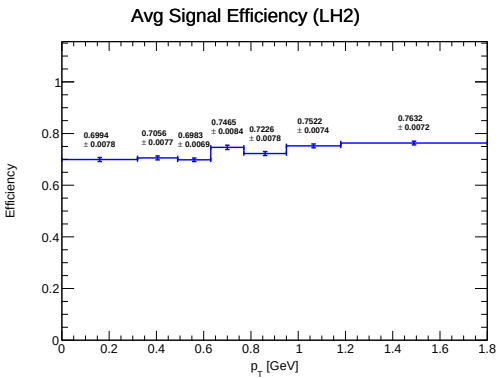


RS57: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS57 LH2)

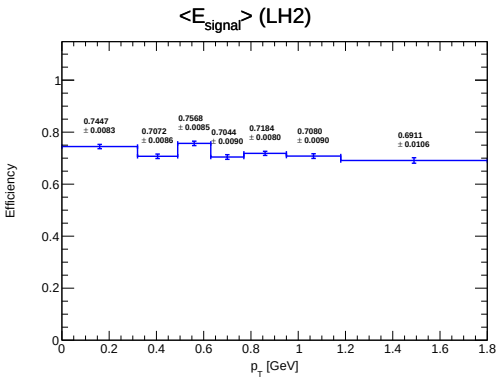


Existing Efficiency (RS57 LH2)

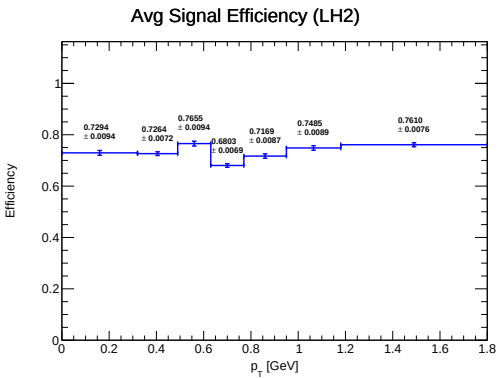


RS59: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS59 LH2)

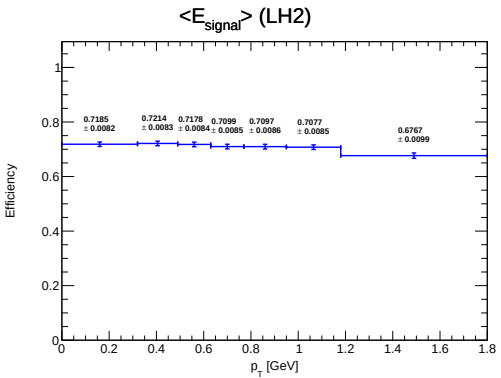


Existing Efficiency (RS59 LH2)

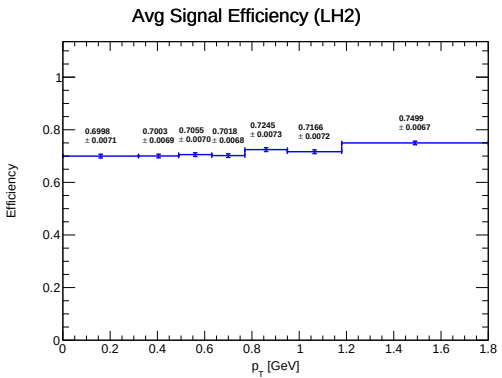


RS62: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS62 LH2)

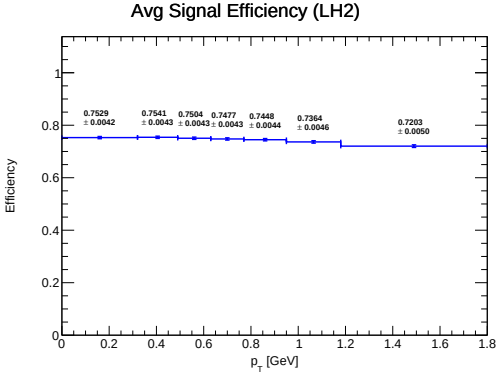


Existing Efficiency (RS62 LH2)

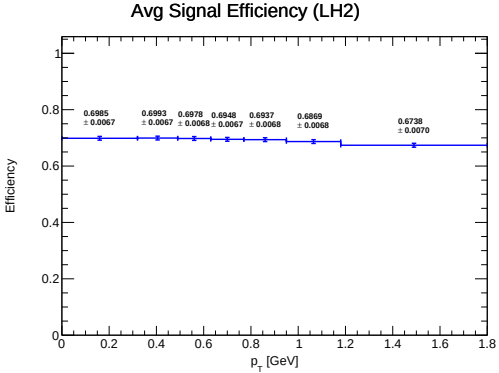


RS67: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS67 LH2)

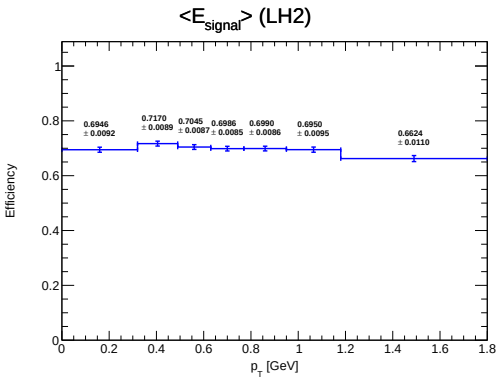


Existing Efficiency (RS67 LH2)

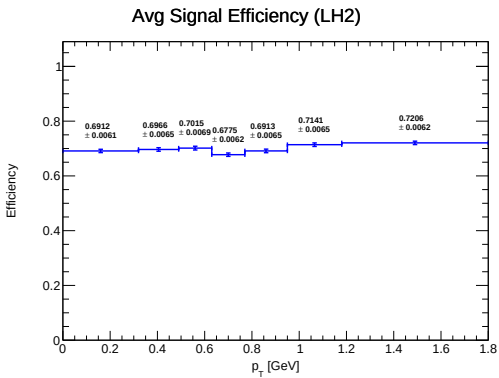


RS70: Latest Vs Existing Reconstruction Efficiencies (LH2)

Latest Efficiency (RS70 LH2)

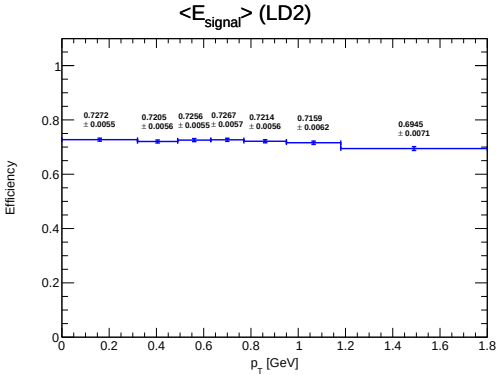


Existing Efficiency (RS70 LH2)

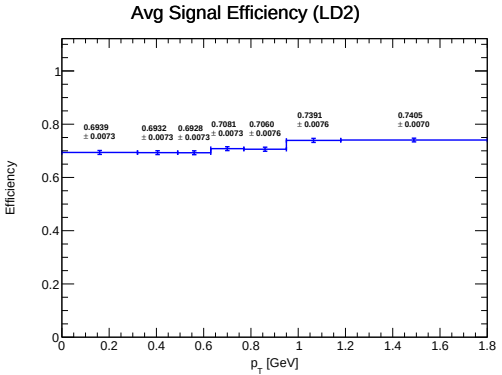


RS57: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS57 LD2)

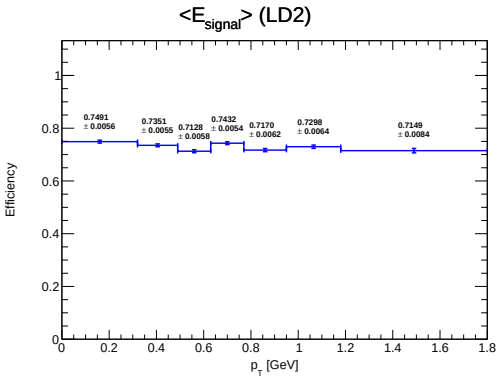


Existing Efficiency (RS57 LD2)

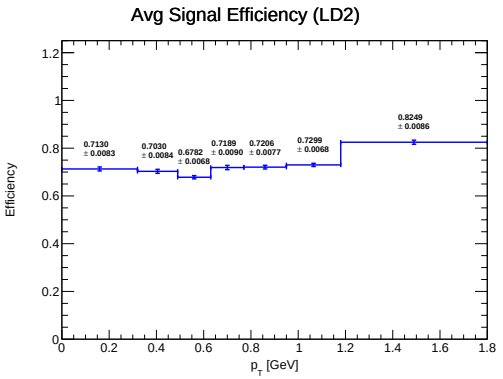


RS59: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS59 LD2)

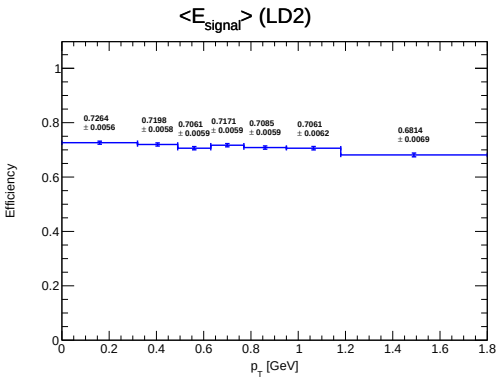


Existing Efficiency (RS59 LD2)

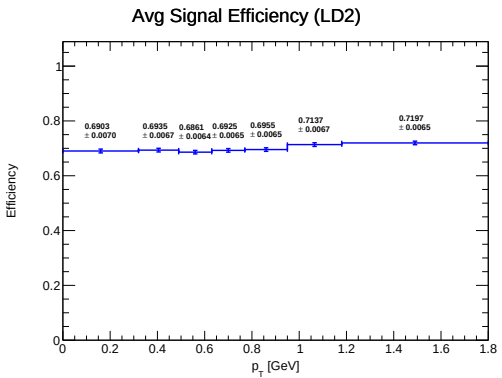


RS62: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS62 LD2)

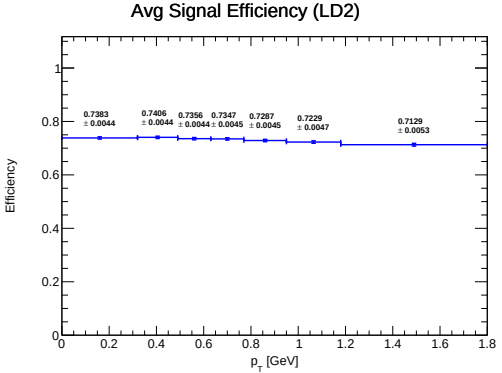


Existing Efficiency (RS62 LD2)

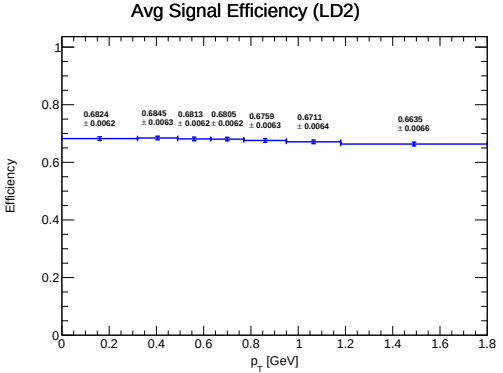


RS67: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS67 LD2)

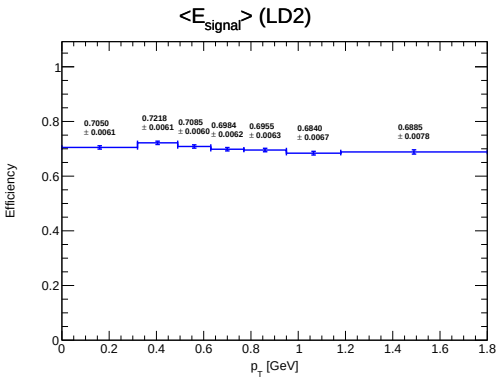


Existing Efficiency (RS67 LD2)

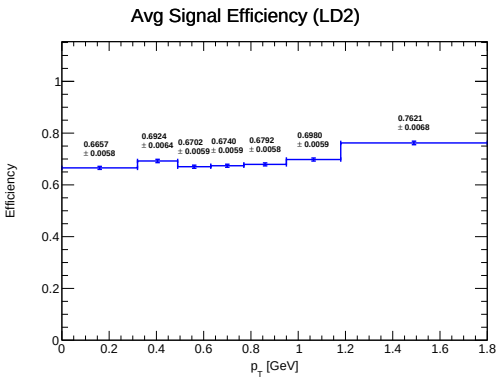


RS70: Latest Vs Existing Reconstruction Efficiencies (LD2)

Latest Efficiency (RS70 LD2)

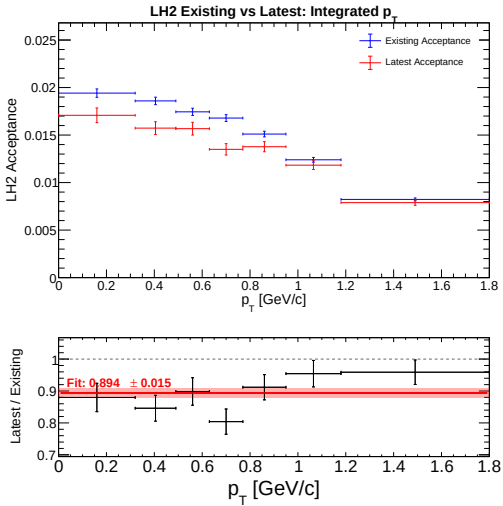


Existing Efficiency (RS70 LD2)

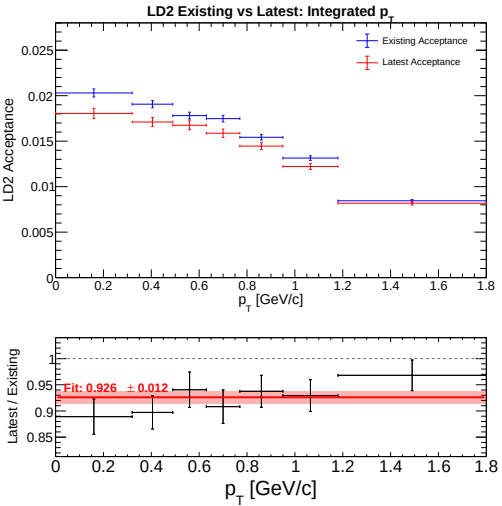


RS57 Acceptance Corrections

LH2 (RS57)

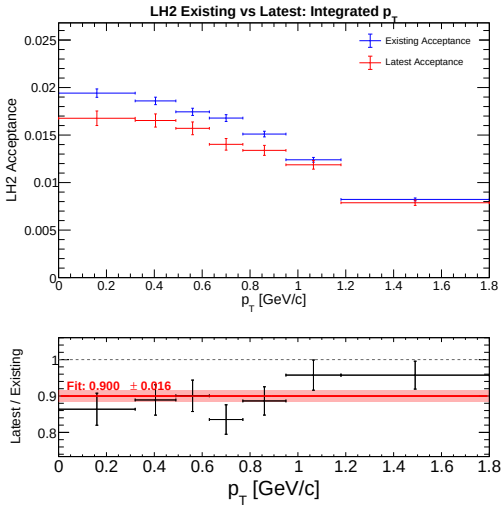


LD2 (RS57)

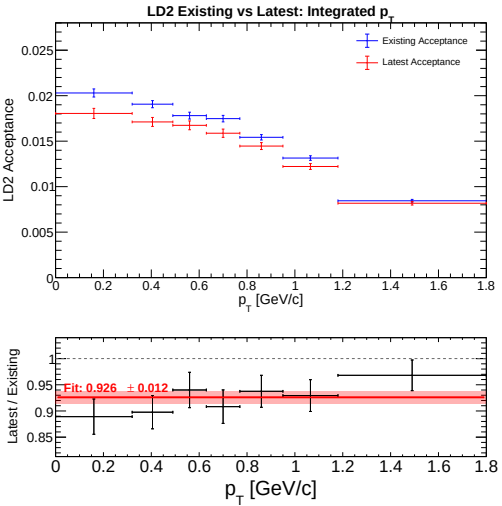


RS59 Acceptance Corrections

LH2 (RS59)

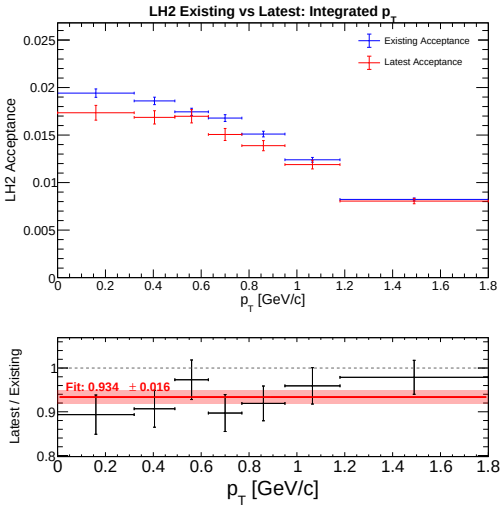


LD2 (RS59)

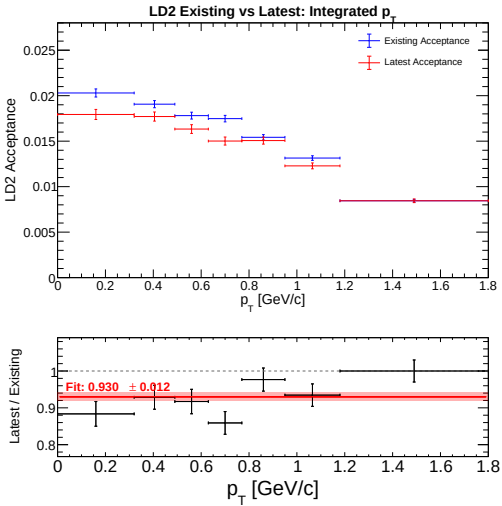


RS62 Acceptance Corrections

LH2 (RS62)

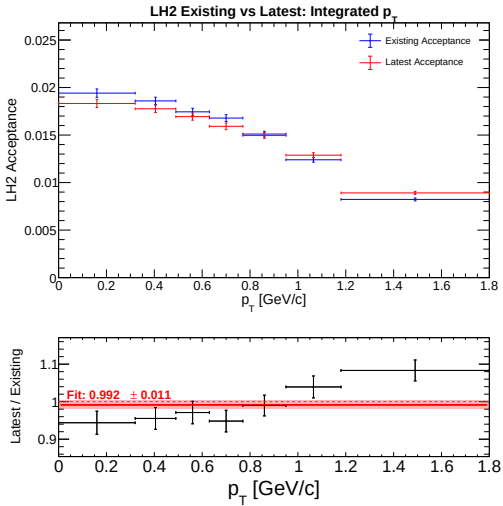


LD2 (RS62)

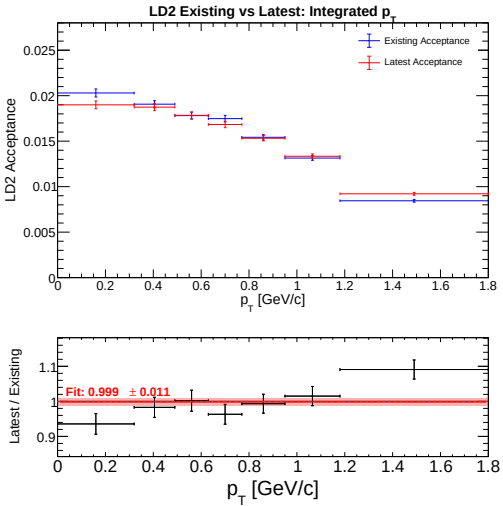


RS67 Acceptance Corrections

LH2 (RS67)

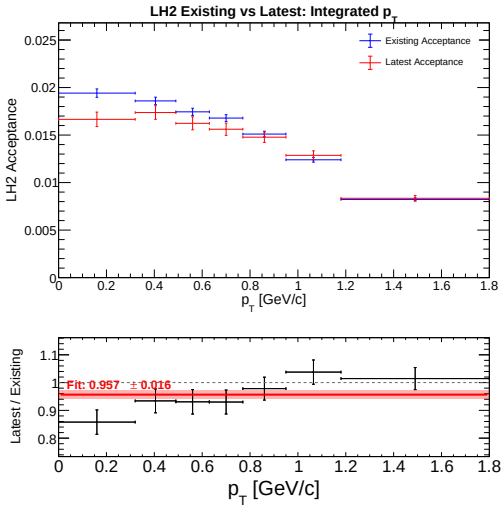


LD2 (RS67)

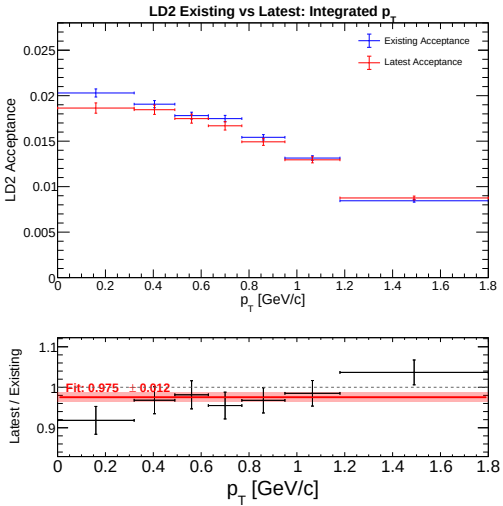


RS70 Acceptance Corrections

LH2 (RS70)

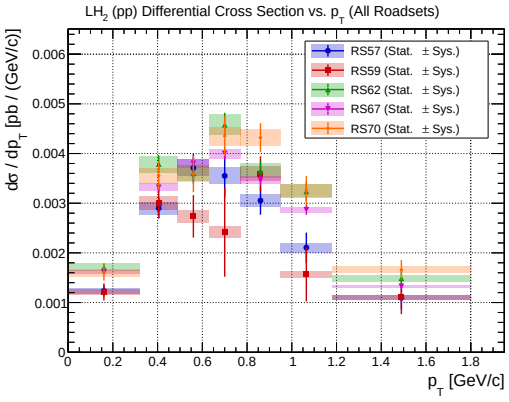


LD2 (RS70)

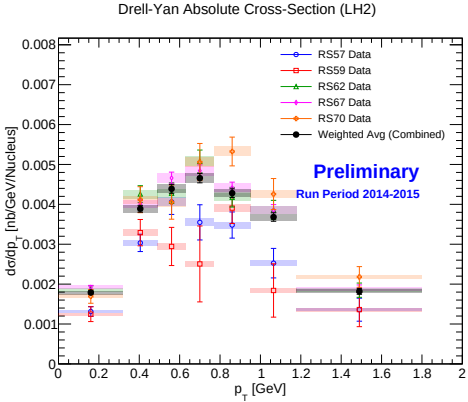


LH2 Cross-Section Comparison All Roadsets

Latest (LH2 All Roadsets)

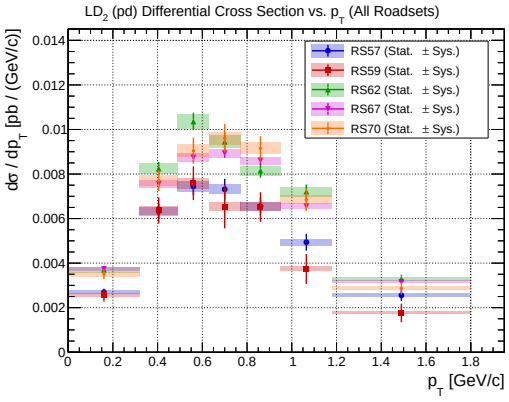


Existing (LH2 All Roadsets)



LD2 Cross-Section Comparison All Roadsets

Latest (LD2 All Roadsets)



Existing (LD2 All Roadsets)

